Dive Buddy Alert Band

Stay in touch. Know where to look.

A reef draws one diver's attention. Their buddy turns to share something else. DBA proposes a private call and a clear acknowledgement, with buddy location supplied by a shared acoustic reference. Two bands, reference equipment and sealed charging form a system for operator-managed boat dives.

A private call. A useful view.

The concept combines a paired attention request with proposed horizontal bearing, relative depth and a shared-position map. Dedicated Dive and Boat views keep elapsed time, motion and the reference position in context.

Direction and map views require the proposed acoustic positioning reference, plus depth and heading sensing on the bands. All interface readings shown here are simulated. DBA remains an aid to buddy procedures, never a safety guarantee.

A shared dive. Two different points of view.

The moment before visual communication.

Underwater, a hand signal depends on being seen. A buddy can be nearby yet outside the other diver's field of view, above the reef or a few metres deeper. Getting their attention is the first step. Understanding where to look is the next.

One press starts the conversation.

Pair the bands before entering the water. A deliberate press requests a vibration on the paired band. The receiving diver presses to acknowledge. The caller sees that human response separately from the band's automatic delivery receipt.

Distance gains context.

The proposed navigation view combines separation with horizontal bearing and a clear shallower-or-deeper cue. A second view places both divers on a horizontal grid. Dive information and the boat-associated reference each have a dedicated screen, so every number has a purpose.

The intended use is a routine buddy connection during a shared dive. The band does not replace visual contact, established separation procedures or a dive computer.

Illustrative scenario. No customer testimonial, founder incident, sales result or demonstrated underwater performance is claimed.

The complete system

Two bands, an acoustic reference and inductive charging.

Two paired bands

Each diver wears a band with a top button, haptic alert and proposed display. Each band measures direct peer range and would sense depth and wrist heading.

Boat-associated acoustic reference

A proposed spaced receiver array and orientation reference supply shared horizontal positions. Placement, calibration, coverage and supported pair count need definition.

Inductive charging dock

The operator charges the sealed modules between uses. Dock capacity, alignment, charge time and reference power arrangements need definition.

Which capabilities depend on the reference?

With a current direct peer link
Paired call, device receipt, human acknowledgement and scalar separation. No horizontal bearing follows from range alone.

With current shared reference fixes
Buddy bearing, horizontal map and boat-relative position. Relative depth also needs current exchanged pressure readings. Wrist heading determines display orientation.

When positioning is unavailable
Clear position-dependent arrows and markers. Keep independently valid direct range and local depth/time. Peer-link loss also clears exchanged buddy data.

The proposed first customer

Operator-managed recreational boat dives

A boat-based operator could manage the acoustic reference and charging equipment, allocate paired bands before departure and support their use across successive trips. That makes the added equipment part of one managed service.

Proposed first market. Customer demand has not been established.

Before entry

Operator assigns the buddy pair, confirms charge, deploys the reference and checks readiness. Divers confirm pairing and adjust their straps.

During the dive

Divers use the top button to call and acknowledge. Valid position information supports a glance toward the buddy. Established buddy procedures continue.

Between trips

Operator collects the bands and manages cleaning, charging, inspection and service. Time per turnaround and staffing burden remain to be established.

The purchaser and the wearer

The operator would own and manage the system. Divers would wear paired bands during the trip. An equipment purchase with a possible rental or included-service offer is a commercial hypothesis. Setup burden, service obligations and willingness to pay still need customer evidence.

First-product priorities

The proposed release scope and the wider concept.

First-product essentials

Pair identity, call, explicit human acknowledgement, clear missing-data states, adjustable retention and sealed charging. These define the basic interaction and daily handling.

Navigation feasibility decision

Buddy bearing and relative depth are the proposed first-product location view. Include them only if the reference architecture can meet agreed coverage, accuracy, latency, power and setup constraints.

Later interface options

Map, Boat, additional motion statistics and orientation comparisons remain in the full concept demonstration. Their release priority follows the core interaction and location decision.

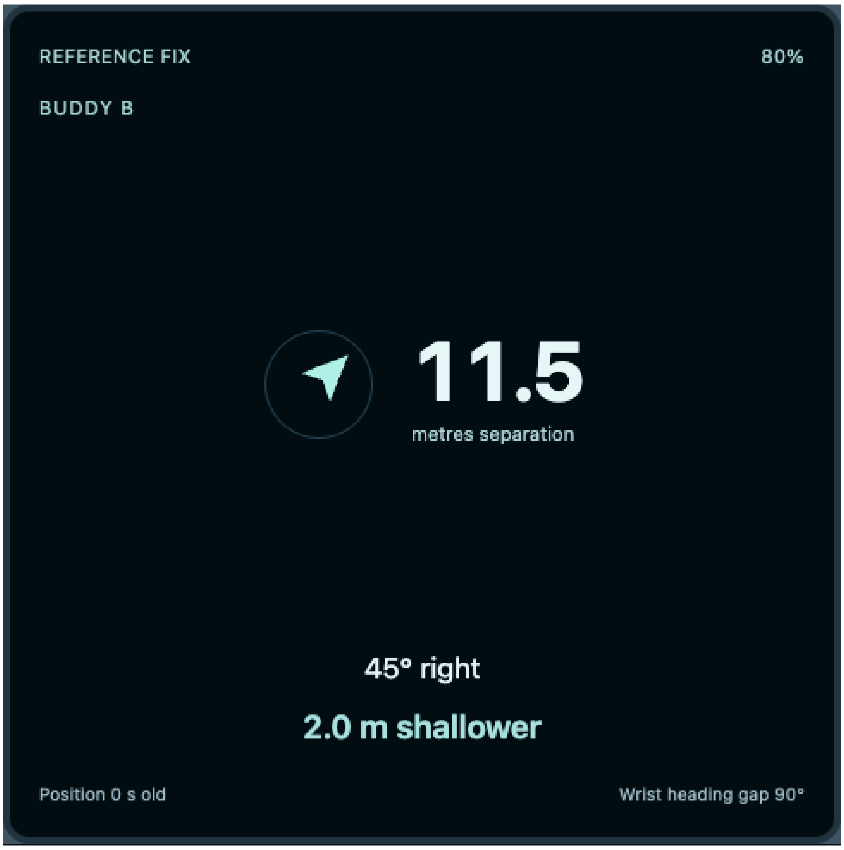
A call-and-range-only product would need a separate commercial decision if reference-assisted navigation proves unsuitable. It is not an automatic substitute for the proposed navigation product.

The demonstration includes all four views

Buddy location is the proposed priority view. Map, Dive and Boat remain available to explain the broader vision. Their presence in the demonstration does not commit them to the first release. A development partner should recommend scope against the agreed use case, cost and technical constraints.

Buddy bearing and relative depth

A horizontal direction with a separate vertical relationship.



11.5 metres is the separation. The arrow and wrist-relative angle locate the buddy horizontally. "2.0 m shallower" supplies the depth relationship. North-up mode expresses the bearing as a compass direction and angle.

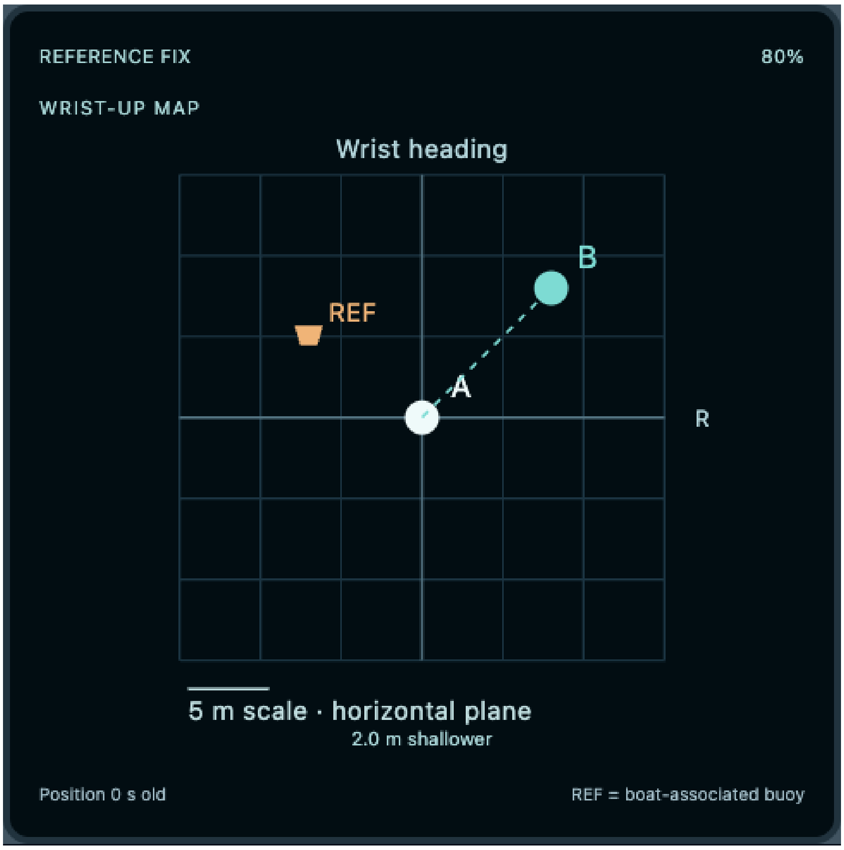
An arrow needs more than range

Direct ranging still uses one wrist transducer. The proposed boat-associated acoustic reference supplies shared horizontal positions; pressure and heading sensing are added to each band. Heading is the wrist reference, which may differ from swimming direction.

Same horizontal position clears the directional arrow. A shallower/deeper cue describes where a buddy is, not an instruction to ascend or descend. All example readings are synthetic.

Both divers on one map

A horizontal plane, with depth kept explicit.



The map centres the wearer and places the paired diver and boat-associated reference on a scaled grid. North-up uses east and north axes; wrist-up rotates the view around the wearer's heading.

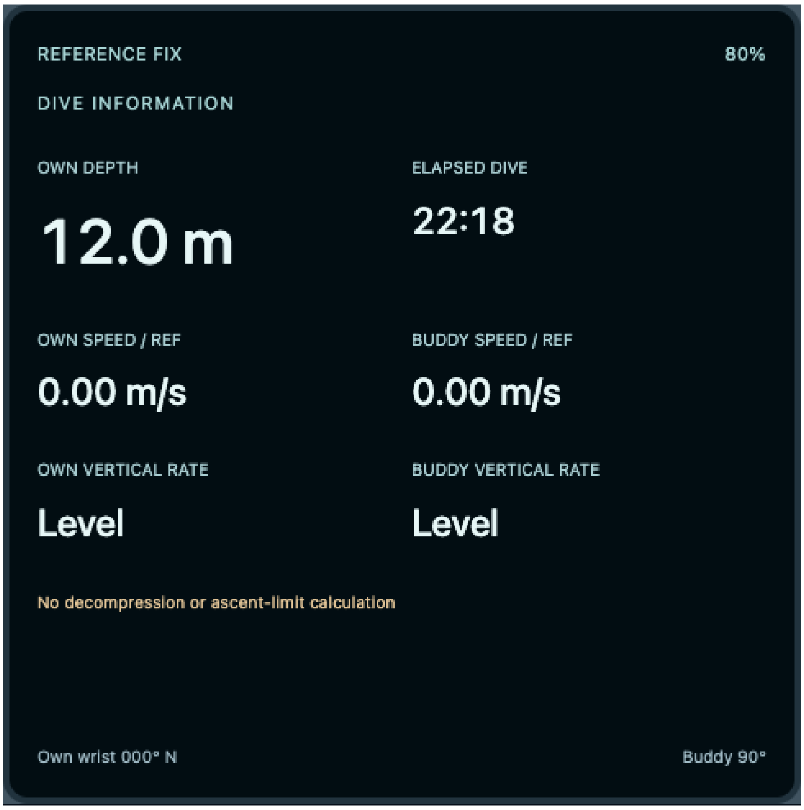
A map point is not a route

The grid represents relative horizontal positions. It does not contain reef geometry, hazards or a safe path. The below-map depth cue remains separate from the horizontal projection. A current peer data link and valid fixes for both divers are required to show the buddy marker.

Position updates can be paused in the demonstration. After three seconds, stale position markers and arrows clear. That illustrative age limit does not establish a real sensing update rate or accuracy.

Time, depth and movement

Each measure answers a different question.



Dive information shows own depth, elapsed dive time, both reference-relative horizontal speeds, both vertical rates and wrist headings. The moving and ascending scenarios make those distinctions visible.

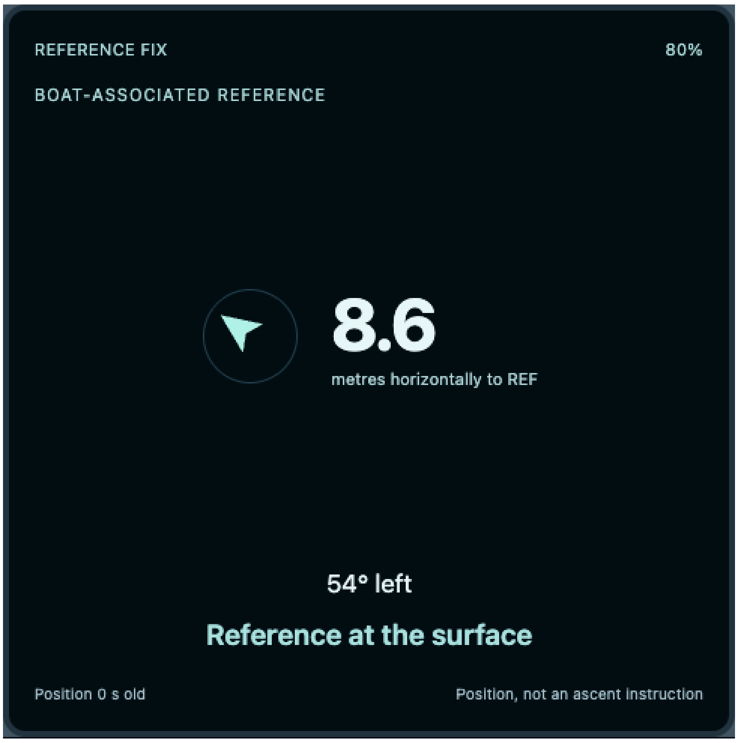
Horizontal speed is relative to the reference, not speed through water. Vertical rate is derived from depth change over time. Neither is a decompression calculation. An ascent-rate readout cannot establish prevention of decompression sickness.

Local depth and time remain when the acoustic reference is unavailable. Exchanged buddy depth and motion require a current peer data link. Reference-derived horizontal speeds clear when their fixes expire.

The example clock starts at 22:18 and can play a 20-second movement sequence. The interface is a simulation, not a diving instrument.

A reference back at the boat

Proposed equipment and the architecture behind navigation.



The Boat view shows horizontal distance and bearing to the associated reference. A reference mounted or deployed near the boat must have a defined relationship to it. A drifting or repositioned reference changes the meaning of that position.

Reference array to shared positions to band displays

A spaced acoustic receiver array and orientation reference would support localization. Band depth and wrist heading complement those positions. This is a proposed architecture: array dimensions, aperture, synchronization, calibration, motion compensation, accuracy and coverage remain undefined.

The trade-off is extra equipment, deployment and power. The reference is represented as a system diagram; this package does not supply invented hardware CAD or claim compatibility with an existing manufacturer's system.

Distance is horizontal to the reference, not a safe ascent instruction. Loss of the reference clears the boat arrow and position.

One physical button

Proposed interaction and independent data availability.

Pair before the dive

Surface pairing establishes the buddy identity. The browser demonstration starts paired. Pair allocation and nearby-pair isolation require firmware definition.

Short press: select a view

At idle, short presses cycle Buddy, Map, Dive and Boat. Web tabs expose those choices directly. The physical interface remains one top button, not a touchscreen.

Hold one second: call

A hold requests three pulses on the paired band. The caller first sees Calling, then device receipt. The receiving diver short presses to acknowledge; only that action establishes a human response.

Cancel and return

A hold cancels an active call. A short press returns from an acknowledged call to the selected view. When the peer link fails, remote cancellation remains unconfirmed.

Data failures

Peer loss clears buddy range and exchanged buddy data. Reference loss or stale positioning clears arrows, map points and reference speeds. Current direct range, own depth and dive time can remain available independently.

Low battery

The demo lowers A to a synthetic 12% and suspends A's positioning. This is proposed UI behavior; it does not establish a real low-battery threshold or reserve.

Adjust the fit. Keep the connection.

Mechanical adjustment and three nominal strap layouts.

A split silicone strap wraps around the wrist or suit cuff. Its overlapping tail adjusts through a proposed acetal buckle, with a silicone keeper securing the loose end. The sealed module remains removable when the strap needs replacing.

Adjustable silicone. Three nominal layouts.

Nominal CAD layouts. Wearer fit ranges are unverified.

Compact, Regular and Cuff are three modeled layouts, with nominal internal reference diameters of 50, 62 and 78 mm. Their circular reference circumferences are about 157, 195 and 245 mm. These dimensions describe CAD layouts; they are not established wrist-size ranges.

An overlapping silicone tail has ten modeled adjustment holes, with a buckle envelope and tail keeper. The main strap is 24 mm wide and 4 mm thick; the tail is 22 mm wide and 2.4 mm thick.

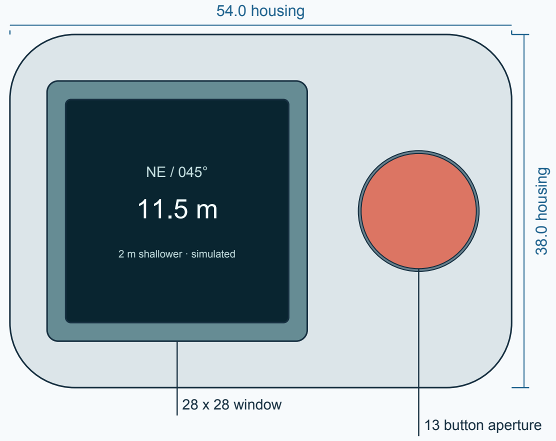
Corrosion resistance is a design requirement. The proposed exterior combines silicone, an acetal buckle with nonmetal retention, and a polymer housing and window. No exposed steel pin or charging contacts are specified. Material grades, seal durability, buckle strength and the usable fit range remain unvalidated.

The current assembly

Rev F nominal dimensions, derived from the parametric source.

DISPLAY MODULE / TOP VIEW

Window and top button in the current parametric assembly.



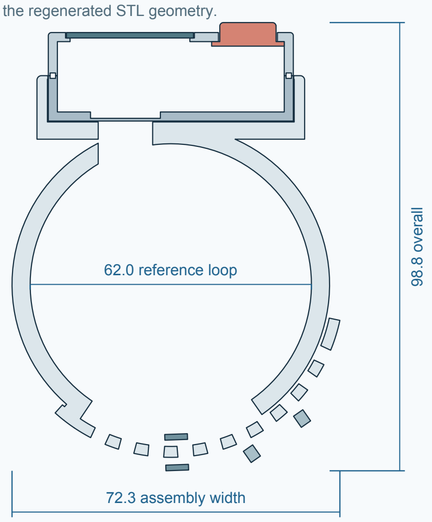
24 x 24 active display area

Screen and window sizes are concept parameters; readability is unverified.

DBA / REV F | NOMINAL mm | CONCEPT GEOMETRY - NOT A MANUFACTURING RELEASE

DISPLAY ASSEMBLY / FRONT SECTION

Centre section from the regenerated STL geometry.



Band width 24 / radial wall 4 / cradle width 43

DBA / REV F | NOMINAL mm | CONCEPT GEOMETRY - NOT A MANUFACTURING RELEASE

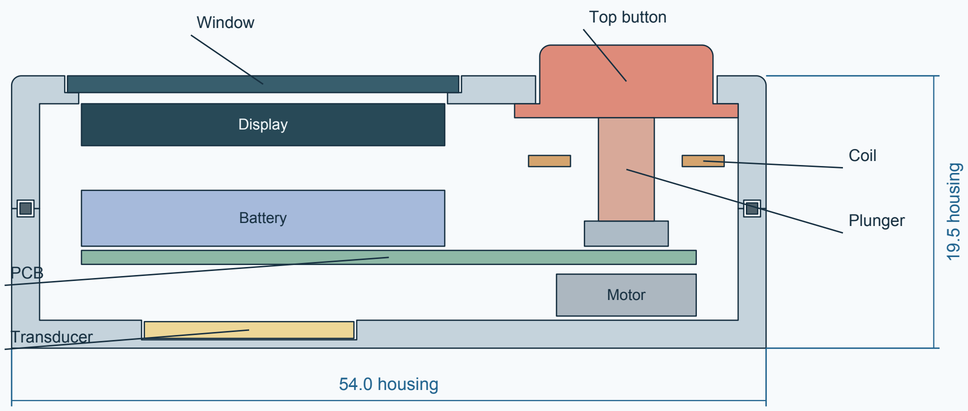
54 x 38 x 19.5 mm housing, excluding 2.2 mm button protrusion. Regular assembly height: 98.8 mm. 18 component envelopes. Compact and Cuff variants move the same module onto different reference strap layouts.

Inside the sealed module

Display, pressure sensing, heading sensing and charging.

DISPLAY MODULE / CENTRE SECTION

Current STL section. Color identifies component envelopes.



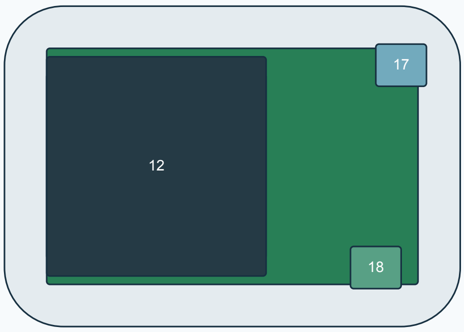
Coil is beneath the button side of the cap; the plunger passes through its centre opening.

Nominal clearances are modeled. Wiring, supports, sealing and charging efficiency remain undefined.

DBA / REV F | NOMINAL mm | CONCEPT GEOMETRY - NOT A MANUFACTURING RELEASE

INTERNAL LAYOUT / CAP REMOVED

Plan projection of selected component envelopes; depths differ.



05 PCB · 07 Battery · 12 Display · 17 Pressure sensor · 18 Heading / inertial sensor

Pressure coupling, wiring, mounting and actual component selections remain undefined.

The external acoustic reference is a system proposal; no fabricated array CAD is supplied.

DBA / REV F | NOMINAL mm | CONCEPT GEOMETRY - NOT A MANUFACTURING RELEASE

Display body: 26 x 26 x 3 mm. Window: 28 x 28 x 1.2 mm. Active area: 24 x 24 mm. Sensor parts are envelopes, not selected devices. Pressure coupling, wiring, supports, seals, optical stack and manufacturing tolerances remain undefined.

Feasibility decision criteria

P0 decisions define the first-product scope. P1 decisions follow that scope. Units specify how requirements should be expressed; they are not measured results or approved acceptance limits.

P0 / Intended use and operating depth

Define the supported dive environment and depth rating in metres, with temperature and salinity bounds. The old 50 m proposal is unverified.

Responsible: Systems / mechanical
Decision: Operating limits agreed before detailed enclosure design.

P0 / Positioning and coverage

Specify horizontal and relative-depth error in metres, bearing error in degrees, usable reference coverage in metres and maximum supported pairs. No current numeric targets.

Responsible: Underwater acoustics / estimation
Decision: A credible reference architecture for the stated use case.

P0 / Message and data quality

Retain 20 m as the direct-range target. Define call delivery and acknowledgement delay in seconds, update interval, stale-data cutoff and missed/false alert limits. Demo timings are illustrative.

Responsible: Firmware / acoustics
Decision: Fresh, stale, unavailable and human-response states remain distinct.

P0 / Energy and turnaround

Define band and reference runtime in hours, reserve, charge time and operator turnaround in minutes. Include display and navigation duty cycles.

Responsible: Electronics / operations
Decision: Energy and charging fit the proposed trip schedule.

P0 / Display and control usability

Define readable distance, light and viewing-angle conditions for the 24 x 24 mm active area, glove actuation and button force in newtons, and perceptible haptic patterns through specified suit thicknesses.

Responsible: Human factors / industrial design
Decision: A diver can identify the call, response and location state with the intended equipment.

Feasibility decision criteria

P0 decisions define the first-product scope. P1 decisions follow that scope. Units specify how requirements should be expressed; they are not measured results or approved acceptance limits.

P0 / Fit, sealing and materials

Define wrist/cuff coverage in millimetres, adjustment and retention force in newtons, and material/seal durability requirements. Nominal strap layouts do not establish fit ranges.

Responsible: Mechanical / materials
Decision: The wearable meets agreed comfort, retention and saltwater requirements.

P0 / Operator deployment

Define array placement, calibration, reference motion handling, setup time in minutes and pair allocation. The reference's relationship to the boat must remain explicit.

Responsible: Systems / operations
Decision: Reference handling fits an operator's working routine.

P0 / Commercial envelope

Set system selling-price or rental-model assumptions, target margin, order volume, development budget and milestone dates. No current values are approved.

Responsible: Rafael / commercial lead
Decision: Feasibility scope and commercial limits are agreed before a full development commitment.

P1 / Later views and service

Rank map, boat, motion and orientation views. Define battery/module service, replacement strap supply and lifecycle obligations.

Responsible: Product / service / firmware
Decision: Optional features and ongoing support have explicit scope and cost ownership.

Engineering specification

Register 1-5 of 24. Proposed behavior, nominal dimensions and unresolved implementation.

Product scope / First-product proposal

Paired attention, acknowledgement and reference-assisted buddy navigation. An aid to buddy procedures, never a safety guarantee.

Engineering notes: No demonstrated underwater performance, safe-route guidance, reunion or decompression protection. Core interaction first; navigation is conditional on feasibility. Map, Boat and additional statistics are lower-priority interface options.
Discipline: Systems / product

Form and service / Design decision

Adjustable silicone strap and keeper; proposed acetal buckle with nonmetal retention; sealed removable module and one silicone top button.

Engineering notes: No exposed steel pin is specified. Buckle latch, attachment, retention and service details remain undefined. Strap replacement does not make the sealed module user-openable.
Discipline: Mechanical / industrial design

Charging interface / Design decision

Inductive charging through the sealed cap; no band charging port or exposed charging contacts.

Engineering notes: Receiver coil is beside the display and around the button plunger. Dock alignment, ferrite, thermal limits and charging efficiency require definition.
Discipline: Electronics / mechanical

Direct acoustic range / Baseline architecture

Approximately 65 kHz two-way time-of-flight with one transducer on each wrist.

Engineering notes: A direct scalar range cannot produce bearing. Transducer, directivity, drive levels, receiver sensitivity and protocol remain unselected.
Discipline: Underwater acoustics

Range and timing / Historical targets

20 m direct-range target. Rev A assumed approximately ±0.3 m range performance, 1 Hz exchange and fixed 2 ms reply delay.

Engineering notes: Unmeasured targets; none establishes reference-position accuracy, coverage or update rate. Timing precision alone does not establish accuracy.
Discipline: Acoustics / firmware

Engineering specification

Register 6-10 of 24. Proposed behavior, nominal dimensions and unresolved implementation.

Operating envelope / Unverified

Rev A proposed 50 m operating depth. The current concept has no pressure rating, validated runtime or assembled mass.

Engineering notes: Immersion, temperature, salinity and pressure convention require definition. Earlier four-dive runtime and 28 g mass do not carry over to Rev F.
Discipline: Systems / mechanical

Pairing / Proposed behavior

Surface-only pairing identifies the buddy before the dive. The demonstration begins paired.

Engineering notes: Pairing gesture and address allocation need definition; nearby-pair isolation, authentication and reacquisition are unresolved.
Discipline: Firmware

Call and acknowledgement / Proposed behavior

Hold the top button for 1 s to call. The receiver feels three pulses; a short press acknowledges. Device receipt and human acknowledgement are separate states.

Engineering notes: Web call buttons simulate the hold. Retry limits, delayed packets, simultaneous calls and timeout behavior need protocol definition.
Discipline: Firmware / interaction

View selection and cancel / Proposed behavior

At idle, short presses cycle Buddy, Map, Dive and Boat. Incoming calls interrupt the view. A hold cancels an active call; an acknowledged call returns to the view with a short press.

Engineering notes: Web tabs expose view selection directly. Gesture arbitration, glove actuation and remotely confirmed cancellation remain unresolved.
Discipline: Firmware / interaction

Haptic vocabulary / Revised proposal

Three pulses request attention. Optional demonstration distance pulses become more frequent at shorter range.

Engineering notes: Demo cadence is illustrative, not the old Rev A release waveform. Amplitude, suit transmission, hysteresis and distinguishing alerts require definition; close does not mean reunited.
Discipline: Interaction / firmware

Engineering specification

Register 11-15 of 24. Proposed behavior, nominal dimensions and unresolved implementation.

Missing and stale data / Proposed behavior

Loss of peer range clears buddy range and exchanged data. Loss or age-out of the reference fix clears arrows, map positions and boat position. Independent depth and time remain.

Engineering notes: Demo uses a 3 s position-age limit and a synthetic 20 m cutoff per acoustic leg. These illustrate failures; actual quality thresholds, latency, uncertainty and outlier rejection remain unestablished.
Discipline: Systems / firmware

Acoustic position quality / Open design work

A spaced reference receiver array and orientation reference would estimate shared horizontal positions from acoustic exchanges.

Engineering notes: Array geometry, clock synchronization, calibration, motion compensation, near-vertical geometry and multipath rejection need design. No locator hardware or measured accuracy is supplied.
Discipline: Underwater acoustics / estimation

Power budget / Pending estimate

Display, depth sensing, heading sensing and reference-assisted exchanges add loads to the original band concept.

Engineering notes: 150 mAh / 3.7 V remains a historical reference, not selected capacity. New band and reference power budgets, reserve and charging time are required.
Discipline: Electronics

Battery behavior / Proposed behavior

Local battery remains visible. The demo reduces A to 12% and suspends A's positioning; peer range and local dive data remain.

Engineering notes: 12% is a synthetic UI state, not a validated threshold or reserve. Cell protection, cold-water behavior and battery life are unestablished.
Discipline: Electronics / firmware

Housing, seals and corrosion resistance / Design requirement / nominal geometry

Saltwater corrosion resistance is required. Proposed polymer housing and nonmetal window; internal electronics isolated from seawater. Housing: 54 x 38 x 19.5 mm, plus 2.2 mm button protrusion; 2 mm shell wall.

Engineering notes: Resin grades, seal interfaces, pressure-sensor water coupling and acoustic membrane require design. Saltwater durability, UV resistance and pressure performance are unvalidated; no corrosion-proof rating is claimed.
Discipline: Mechanical / acoustics

Engineering specification

Register 16-20 of 24. Proposed behavior, nominal dimensions and unresolved implementation.

Internal packaging / Rev F nominal layout

18 envelopes include display, window, plunger, adjustment buckle, keeper, pressure sensor and heading/inertial sensor.

Engineering notes: Nominal non-intersection does not establish wiring, PCB routing, sensor pressure port, supports, actual components or fastener design. Reference array remains a system diagram, without fabricated CAD detail.
Discipline: Mechanical / electronics

Strap and button / Rev F nominal geometry

Compact / Regular / Cuff reference diameters: 50 / 62 / 78 mm. Main strap: 24 mm wide, 4 mm thick. Overlap tail: 22 mm wide, 2.4 mm thick, ten 3.5 mm holes.

Engineering notes: 157 / 195 / 245 mm circular circumferences are reference layouts, not wearer ranges. Proposed UV-stabilized acetal buckle grade, creep, latch strength, silicone formulation, safe extension and suit compression remain undefined.
Discipline: Industrial design / mechanical

Drawing authority / Source-derived dimensions

Regular assembly is 72.3 mm wide and 98.8 mm high. All three size variants derive from dba_cad.py.

Engineering notes: Rev F drawings and meshes represent current geometry. Prior revisions are historical. Dimensions describe envelopes, not manufacturing readiness.
Discipline: Mechanical

Commercial case / Proposed first market

Operator-managed recreational boat dives. An operator-owned kit includes paired bands, inductive charging and a boat-associated acoustic reference.

Engineering notes: Market and operating-model hypothesis. DBA demand, willingness to pay, setup/turnaround burden and supported pair count remain unestablished. See the Rev G commercial brief.
Discipline: Product / commercial

Economics / Current estimate pending

No current complete-system price, target margin, volume forecast, approved development budget or committed launch date.

Engineering notes: Rev A desk estimates are historical and do not price the navigation system. The partner brief defines the cost coverage and decisions required.
Discipline: Commercial / manufacturing

Engineering specification

Register 21-24 of 24. Proposed behavior, nominal dimensions and unresolved implementation.

Project status / Presentation Rev G / CAD Rev F

Concept by Rafael. Current files contain four interactive navigation views and three parametric adjustable strap layouts.

Engineering notes: Model consistency and browser behavior are verified separately from physical performance. No parts procurement or physical test program is commissioned.
Discipline: Project lead

Display and readability / Rev F envelope

Display body 26 x 26 x 3 mm; window 28 x 28 x 1.2 mm; active area 24 x 24 mm.

Engineering notes: Enlarged screen studies do not prove legibility at actual scale. Display technology, resolution, contrast, viewing angle and optical stack remain unselected.
Discipline: Interaction / electronics / mechanical

Bearing, depth and orientation / Proposed sensing

Reference-derived horizontal bearing; pressure-derived relative depth; heading sensing supplies wrist orientation and heading difference.

Engineering notes: Wrist heading is not necessarily swimming direction. Same horizontal position yields no directional arrow. A compass heading alone does not locate a buddy.
Discipline: Systems / acoustics / sensors

Dive time and motion / Proposed information

Dive view shows elapsed time, own depth, both reference-relative horizontal speeds and both vertical rates. Boat view shows horizontal distance and bearing to the associated reference.

Engineering notes: Reference-relative speed is not speed through water. Vertical rate is not a decompression calculation or bends prevention. Boat position assumes a defined relation between boat and reference; neither arrow nor depth cue is an ascent instruction.
Discipline: Firmware / estimation / interaction

What published diver feedback adds

Published observations informed the revision; no direct interviews were conducted.

Attention is useful; direction adds utility

The Scuba Diver Life Buddy-Watcher review discusses the value of an addressed underwater alert and the lack of directional information. DBA Rev F responds with a proposed positioning architecture, not an assertion that a simple alert band already provides bearing.

https://scubadiverlife.com/gear/gear-review-buddy-watcher/

An alert still depends on the wearer

Instructor commentary at Different Dive describes using a paired alert device in practice. DBA separates device receipt from the wearer's explicit acknowledgement and leaves cuff transmission and glove handling as human-factors questions.

https://differentdive.com/week-26-i-test-the-buddy-watcher/

Positioning has physical constraints

Garmin's SubWave limitations document describes environmental and geometry constraints for its own positioning system. It is evidence that acoustic navigation needs quality handling, not a transfer of Garmin specifications or compatibility to DBA.

https://www.garmin.com/en-US/legal/subwavedisclaimer/

Vertical rate is not decompression protection

Divers Alert Network discusses ascent rates within broader decompression considerations. DBA provides a proposed rate display without claiming a safe ascent limit, a decompression model or prevention of decompression sickness.

https://dan.org/alert-diver/article/ascent-rates/

The commercial proposition

A private buddy call with readable spatial context.

A private buddy call with readable spatial context, offered through a dive operator that manages the supporting equipment.

Existing buddy procedures

DBA supplements visual communication and the pair's established procedures. The product must justify adding equipment to an already essential routine. The claimed benefit is an addressed prompt before visual contact. No reduction in incidents or separation has been demonstrated.

Paired underwater alert devices

Published Buddy-Watcher reviews provide a relevant comparison for the attention use case. DBA proposes explicit human acknowledgement and reference-assisted spatial context. Superiority, novelty and purchase preference are unproven.

Integrated underwater communication and positioning

Garmin publishes limitations for its SubWave system, providing a reference for environment and positioning constraints. A dedicated operator-managed system must justify its equipment burden and total price against integrated alternatives. No compatibility or feature-parity claim is made.

Documented
A reproducible concept model, dimensions, simulated interactions, failure-state behavior and published third-party product commentary.

Still needed for a business decision
DBA-specific diver and operator feedback, frequency and importance of the use case, willingness to pay, equipment-handling burden and comparison with alternatives.

Commercial decision
Proceed only if the intended benefit and acceptable system price justify the operator's setup, turnaround and service obligations. No demand threshold or purchase commitment is established.

Published source references appear on the preceding page. They concern other products and do not establish demand, superiority, compatibility or endorsement of DBA.

Commercial inputs for a scoped proposal

Current system: paired bands, reference equipment and charging.

Purchaser and offer
Proposed operator-owned kit: paired bands, acoustic reference and inductive charging. Supported pair count and shared-reference capacity remain open.

Revenue model
Operator equipment purchase with a possible diver rental or included-service model. Proposed options, not established customer preferences.

System price and margin
No current target, quote or margin established. Define the acceptable complete-system price and replacement-band price with the intended purchaser.

Volume
Initial operator count, kits per operator, bands per kit and replacement demand are undecided. Rev A's 5,000-pair assumption is not a forecast.

Development budget
Not supplied. Rafael and the partner must agree a budget for a bounded feasibility/scoping engagement before commissioning development.

Schedule
No launch date or development duration committed. Request proposed milestones, dependencies and decision points with the scope.

Cost coverage
Account for bands, reference, docks, engineering, tooling, applicable compliance work, freight, duty, service and warranty. Current full-system costs are unknown.

Decision owners
Rafael owns the product and investment decisions. The development partner proposes technical scope and estimates. Operator evidence informs workflow and price assumptions.

Historical Rev A basis

The display-free design used about $86 per pair at 5,000 pairs, ±30%, and a $179-199 retail target. These desk estimates excluded tooling, engineering, freight and duty. They do not price the current navigation system.

The development-partner request

Feasibility and scope definition.

Assess the proposed system for operator-managed boat dives and recommend a feasible first product, the essential trade-offs and a priced scope for the next development stage.

Product definition

Recommended first-product capabilities, measurable requirements and explicit exclusions, tied to the intended customer and use case.

Architecture assessment

Acoustic-reference approach, sensing, energy and enclosure constraints, with the largest uncertainties and relevant specialist responsibilities.

Commercial scope

Cost drivers, required customer inputs and a milestone-based proposal identifying deliverables, assumptions, fees and schedule dependencies.

Decision recommendation

Proceed, revise or stop, with the reasons, unresolved evidence and conditions for any subsequent commitment.

Specialist capability

The proposed partner should identify responsibility for underwater acoustics and localization, embedded electronics and firmware, pressure sealing, wearable ergonomics and commercial scoping. Relevant acoustic experience is central to the architecture assessment.

Requested next step: feasibility and scope definition. This presentation does not commission procurement, tooling, manufacturing or a physical hardware test program.

Files for presentation and review

Current release: Presentation Rev G / CAD Rev F.

DBA_Partner_Brief_RevG.pdf
One-page customer, system, priority, commercial-input and partnership brief.

DBA_Interactive.html
Primary presentation: four simulated navigation views, actual 3D strap variants, dimensions, engineering register and downloads. The side depth gauge follows page scrolling; it is a presentation effect.

DBA_Product_Overview_RevG.pptx
Editable 14-slide overview with presenter notes. slides.html provides an optional browser slide viewer.

DBA_Design_Package_RevG.pdf
This concept and engineering brief, including proposed system architecture, controls, failure behavior, fit and source references.

DBA_Annotated_Views_RevF.pdf
Current top, front, internal section, sensor layout and three-size fit views. drawings/DBA-001_drawing_sheet.pdf provides the coordinated engineering sheet.

cad/ and cad/fit-variants/
Regular assembly and part STEP/STL files, with Compact and Cuff variants. design-data.json records parameters and envelopes. Reference array architecture has no physical CAD design.

dba_cad.py, render.py and drawing_sheet.py
Change named parameters in the source, regenerate CAD, then regenerate renders, drawings and embedded meshes. Do not hand-edit STEP or STL files.

presentation/ and README.txt
Presentation source, requirements, screen logic and release-generation instructions. All browser mesh variants retain base64 int16 encoding.

legacy/
Original Rev A package and earlier briefs are historical references. Current presentation uses Rev G. CAD and annotated drawings remain Rev F.

Prepared for Rafael

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